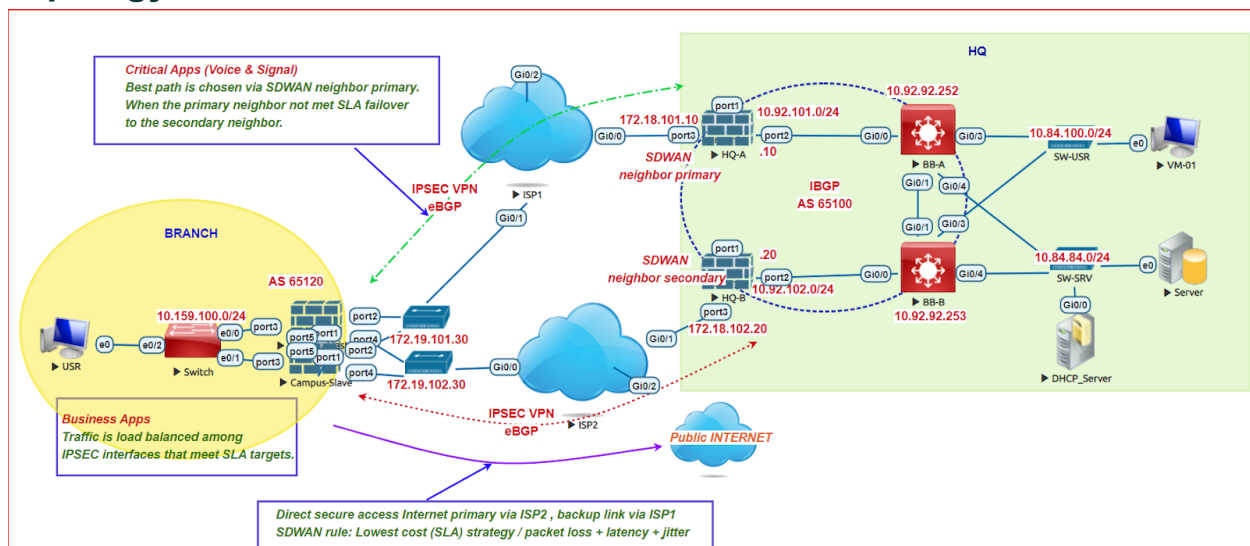


Description

- This Lab describes how to use SD-WAN on FortiGate Firewall control traffic VPN IPSEC and secure INTERNET access.
- A Branch FortiGate has two ISP links for redundancy Internet access.
- At the remote site, two gateways reside in different locations in the Data Center. Two Firewall Gateways connect to Router Backbone using the dynamic routing IBGP.
- Between Branch Firewall and two remote Firewalls at Data Center setup two VPN IPSEC tunnel links for transferring Critical Traffic and Business Traffic.
- Using eBGP routing protocol to exchange the prefixes via VPN IPSEC Tunnel Links.
- DHCP, DNS, RADIUS Servers at Data Center assign IP addresses and manage clients at the Branch site.

Topology



Type of Traffic

No	Type of Traffic	Purpose
1	Critical Apps	- VoIP Signalling SCTP - The traffic for management PC Client as DHCP, DNS, SMTP, RADIUS - Control and Provisioning of Wireless Access Points (CAPWAP)
2	Business Apps	- The rest traffic exchange between Branch and Data Center
3	Secure Internet	- The traffic goes out Public Internet: HTTP, HTTPS, ICMP

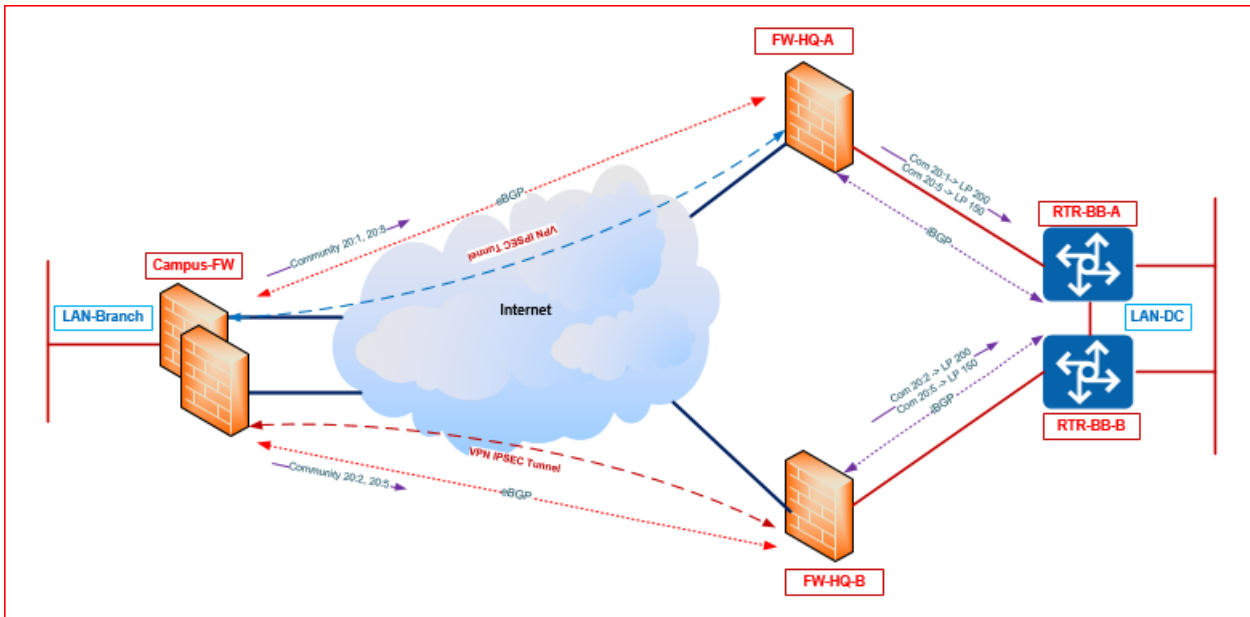
Requirement

1. Critical Apps Traffic are VoIP traffic, signaling traffic as SCTP, CAPWAP, DHCP, DNS, Radius shall transfer via primary IPSEC Tunnel to SD-WAN primary neighbor if it's SLA meets the threshold. And the backup link via secondary IPSEC Tunnel to SD-WAN secondary neighbor. The SLA threshold use three items include packet loss ratio, latency, and jitter.
2. Business Apps Traffic is the rest of Traffic exchange between Branch site and Data Center Site will be load balancer via two links IPSEC Tunnel when they meet the SLA threshold.
3. Secure Internet Traffic uses the primary link via ISP2 and the backup link via ISP1.
4. Marking DSCP 46 (EF) for Critical Traffic and DSCP 34 (AF41) for Business Traffic, other traffic shall be marked to DSCP 0 (BE).
5. A traffic shaping policy applied to the outgoing WAN interface to indicate the priority of Critical_Apps, Business_Apps, and Others.

Solution:

1. Configure BGP routing between the Branch Firewall with two Firewalls at the Data Center site. Use Performance SLA to health check the status of SD-WAN neighbors.

Case	Status	SD-WAN neighbor status	BGP community advertised
Health Check VPN1 link	OK	Primary	To primary: 20:1 To secondary: 20:5
Health Check VPN2 link	OK		
Health Check VPN1 link	OK	Primary	To primary: 20:1 To secondary: 20:5
Health Check VPN2 link	NOK		
Health Check VPN1 link	NOK	Secondary	To primary: 20:5 To secondary: 20:2
Health Check VPN2 link	OK		
Health Check VPN1 link	NOK	Standalone	To primary: 20:5 To secondary: 20:5
Health Check VPN2 link	NOK		



2. Create SD-WAN Rules for Critical Traffic to SD-WAN primary neighbor, backup link to SD-WAN secondary neighbor.
3. Create An SD-WAN Rule for load balancing Business Traffic.
4. Create An SD-WAN Rule for Secure Internet with the primary link via ISP2, the backup link via ISP1.
5. Marking DSCP for the outgoing traffic on SD-WAN Rule follows the table:

Type of Traffic	DSCP marking
Critical_Apps	46
Business_Apps	34
Other	0

6. Apply The Traffic Policy Profile to the outgoing WAN Interfaces

Class of Traffic	Guaranteed Bandwidth	Maximum Bandwidth	Priority
Critical_Apps	90%	100%	critical
Business_Apps	8%	100%	high
Others	2%	100%	low

Configuration Roadmap

1. **Basic configuration on Three Firewalls: HA, Interface, management:**
At The Data Center
 - a. **Create interfaces**

HQ-A # config system interface

```
edit "port2"  
    set vdom "root"  
    set ip 10.92.101.10 255.255.255.0  
    set type physical  
    set alias "Inside"  
    set role lan  
next  
edit "port3"  
    set vdom "root"  
    set ip 172.18.101.10 255.255.255.0  
    set type physical  
    set alias "WAN"  
    set estimated-upstream-bandwidth 2000  
    set estimated-downstream-bandwidth 2000  
    set role wan  
next  
end
```

HQ-B # config system interface

```
edit "port2"  
    set vdom "root"  
    set ip 10.92.102.20 255.255.255.0  
    set type physical  
    set alias "Internal"  
    set role lan  
next  
edit "port3"  
    set vdom "root"  
    set ip 172.18.102.20 255.255.255.0  
    set type physical  
    set alias "WAN"  
    set estimated-upstream-bandwidth 2000  
    set estimated-downstream-bandwidth 2000  
    set role wan  
next  
end
```

b. Create all subnets in Data Center and Branch Sites*HQ-A # config firewall address*

```
edit "LAN-BR"  
    set subnet 10.159.100.0 255.255.255.0
```

```

next
edit "LAN-SRV"
    set subnet 10.84.84.0 255.255.255.0
next
edit "LAN-USR"
    set subnet 10.84.100.0 255.255.255.0
next
edit "INTERNAL_HQ_A"
    set subnet 10.92.101.0 255.255.255.0
next
edit "TUNNEL_ADD"
    set subnet 10.1.1.0 255.255.255.252
next
end
HQ-A # config firewall addrgrp
edit "LAN-HQ"
    set member "LAN-SRV" "LAN-USR" "INTERNAL_HQ_A"
next
edit "VPN_via_BGP"
    set member "INTERNAL_HQ_A" "LAN-BR" "LAN-SRV" "LAN-USR" "TUNNEL_ADD"
next
end

```

HQ-B # config firewall address

```

edit "LAN-BR"
    set subnet 10.159.100.0 255.255.255.0
next
edit "LAN-SRV"
    set subnet 10.84.84.0 255.255.255.0
next
edit "LAN-USR"
    set subnet 10.84.100.0 255.255.255.0
next
edit "INTERNAL_HQ_B"
    set subnet 10.92.102.0 255.255.255.0
next
edit "TUNNEL_ADD"
    set subnet 10.1.1.4 255.255.255.252
next
end

```

HQ-B # config firewall addrgrp

```

edit "LAN-HQ"
    set member "LAN-SRV" "LAN-USR" "INTERNAL_HQ_B"
next

```

```

edit "VPN_via_BGP"
  set member "INTERNAL_HQ_B" "LAN-BR" "LAN-SRV" "LAN-USR" "TUNNEL_ADD"
next
end

```

c. Configure the iBGP routing between two firewalls and Router Backbone

HQ-A # config router bgp

```

set as 65100
set ebgp-multipath enable
config neighbor
  edit "10.92.101.252"
    set next-hop-self enable
    set remote-as 65100
  next
end
config network
  edit 1
    set prefix 10.92.101.0 255.255.255.0
  next
end

```

HQ-B # config router bgp

```

set as 65100
set ebgp-multipath enable
config neighbor
  edit "10.92.102.253"
    set next-hop-self enable
    set remote-as 65100
  next
end
config network
  edit 1
    set prefix 10.92.102.0 255.255.255.0
  next
end

```

d. Advertise the default static route on Firewall HQ-A (primary) with the local-preference value higher than HQ-B

HQ-A # config router route-map

```

edit "LP"

```

```

config rule
  edit 1
    set set-local-preference 120
  next
end
HQ-A (bgp) #config redistribute "static"
set status enable
set route-map "LP"
end

```

At Branch

a. Create interfaces

```

Campus-Master # config system interface
edit "port2"
  set vdom "root"
  set ip 172.19.101.30 255.255.255.0
  set type physical
  set alias "WAN1"
  set estimated-upstream-bandwidth 2000
  set estimated-downstream-bandwidth 2000
  set monitor-bandwidth enable
  set role wan
next
edit "port3"
  set vdom "root"
  set dhcp-relay-service enable
  set ip 10.159.100.254 255.255.255.0
  set type physical
  set alias "Internal"
  set role lan
  set dhcp-relay-ip "10.84.84.100"
next
edit "port4"
  set vdom "root"
  set ip 172.19.102.30 255.255.255.0
  set type physical
  set alias "WAN2"
  set estimated-upstream-bandwidth 2000
  set estimated-downstream-bandwidth 2000
  set monitor-bandwidth enable
  set role wan
next

```

```
end
```

b. Create all Subnets in Data Center and Branch Sites

Campus-Master # config firewall address

```
edit "LAN-BR"
  set subnet 10.159.100.0 255.255.255.0
next
edit "LAN-USR"
  set subnet 10.84.100.0 255.255.255.0
next
edit "LAN-SRV"
  set subnet 10.84.84.0 255.255.255.0
next
edit "INTERNAL_HQ_A"
  set subnet 10.92.101.0 255.255.255.0
next
edit "INTERNAL_HQ_B"
  set subnet 10.92.102.0 255.255.255.0
next
edit "TUNNEL_ADD"
  set subnet 10.1.1.0 255.255.255.248
next
end
```

Campus-Master # config firewall addrgrp

```
edit "LAN-HQ"
  set member "LAN-SRV" "LAN-USR" "INTERNAL_HQ_A" "INTERNAL_HQ_B"
next
edit "VPN_via_BGP"
  set member "INTERNAL_HQ_A" "INTERNAL_HQ_B" "LAN-BR" "LAN-SRV" "LAN-USR"
  "TUNNEL_ADD"
next
end
```

2. Configure Internet access via SD-WAN

At The Data Center

a. Create New SD-WAN Zone name INTERNET

HQ-A # config system sdwan

```
set status enable
config zone
  edit "INTERNET"
```

```

next
end

```

HQ-B # config system sdwan

```

set status enable
config zone
  edit "INTERNET"
  next
end

```

b. Add WAN interface member into INTERNET Zone

HQ-A (sdwan) # config members

```

edit 1
  set interface "port3"
  set zone "INTERNET"
  set gateway 172.18.101.1
next

```

HQ-B (sdwan) # config members

```

edit 1
  set interface "port3"
  set zone "INTERNET"
  set gateway 172.18.102.1
next

```

c. Add the default static route via SD-WAN interface

HQ-A # config router static

```

edit 1
  set distance 1
  set sdwan enable
next
end

```

HQ-B # config router static

```

edit 1
  set distance 1
  set sdwan enable
next
end

```

d. Add Firewall Policy for secure access internet: icmp, http, https*HQ-A # config firewall policy*

```
edit 1
  set name "INTERNET"
  set srcintf "port2"
  set dstintf "INTERNET"
  set srcaddr "all"
  set dstaddr "all"
  set action accept
  set schedule "always"
  set service "ALL_ICMP" "HTTP" "HTTPS"
  set nat enable
next
end
```

HQ-B # config firewall policy

```
edit 1
  set name "INTERNET"
  set srcintf "port2"
  set dstintf "INTERNET"
  set srcaddr "all"
  set dstaddr "all"
  set action accept
  set schedule "always"
  set service "ALL"
  set nat enable
next
end
```

e. Configure Performance SLA to health check Google Server*HQ-A (sdwan) # config health-check*

```
edit "CheckINTERNET"
  set server "8.8.8.8"
  set interval 500
  set probe-timeout 500
  set failtime 5
  set recoverytime 5
  set probe-count 30
  set update-cascade-interface enable
  set update-static-route enable
  set members 1
```

```

config sla
  edit 1
    set latency-threshold 50
    set jitter-threshold 50
    set packetloss-threshold 2
  next
end
next
end

```

HQ-B (sdwan) # config health-check

```

edit "CheckINTERNET"
  set server "8.8.8.8"
  set interval 500
  set probe-timeout 500
  set failtime 5
  set recoverytime 5
  set probe-count 30
  set update-cascade-interface enable
  set update-static-route enable
  set members 1
  config sla
    edit 1
      set latency-threshold 50
      set jitter-threshold 50
      set packetloss-threshold 2
    next
  end
next
end

```

f. Configure SD-WAN rule for access Internet

HQ-A (sdwan) # config service

```

edit 2
  set name "INTERNET"
  set mode sla
  set dst "all"
  set src "LAN-HQ"
  set dscp-forward enable
  set dscp-reverse enable
  config sla
    edit "CheckINTERNET"

```

```

        set id 1
      next
    end
    set priority-members 1
  next
end

```

HQ-B (sdwan) # config service

```

edit 2
  set name "INTERNET"
  set mode sla
  set dst "all"
  set src "LAN-HQ"
  set dscp-forward enable
  set dscp-reverse enable
  config sla
    edit "CheckINTERNET"
      set id 1
    next
  end
  set priority-members 1
next
end

```

At Branch

a. Create New SD-WAN Zone name INTERNET

Campus-Master (sdwan) # config zone

```

edit "INTERNET"
next
end

```

b. Add two WAN interface members into INTERNET Zone

Campus-Master (sdwan) # config members

```

edit 1
  set interface "port2"
  set zone "INTERNET"
  set gateway 172.19.101.1
  set cost 10
next
edit 2
  set interface "port4"

```

```

set zone "INTERNET"
set gateway 172.19.102.1
set cost 5
next

```

c. Add the default static route via SD-WAN interface

Campus-Master # config router static

```

edit 1
    set distance 1
    set sdwan enable
next
end

```

d. Add Firewall Policy for secure access internet: icmp, http, https

Campus-Master # config firewall policy

```

edit 1
    set name "INTERNET"
    set srcintf "port3"
    set dstintf "INTERNET"
    set srcaddr "LAN-BR"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL_ICMP" "HTTP" "HTTPS"
    set nat enable
next
end

```

e. Configure Performance SLA to health check Google Server

Campus-Master (sdwan) # config health-check

```

edit "CheckINTERNET"
    set server "8.8.8.8"
    set interval 500
    set probe-timeout 500
    set failtime 5
    set recoverytime 5
    set probe-count 30
    set update-cascade-interface enable

```

```

set update-static-route enable
set members 1 2
config sla
  edit 1
    set latency-threshold 50
    set jitter-threshold 50
    set packetloss-threshold 2
  next
end
next
end

```

f. Create An SD-WAN Rule allows the primary link via WAN2 and the backup link via WAN1

Campus-Master (sdwan) # config service

```

edit 2
  set name "INTERNET"
  set mode sla
  set dst "all"
  set src "all"
  set dscp-forward enable
  set dscp-reverse enable
  config sla
    edit "CheckINTERNET"
      set id 1
    next
  end
  set priority-members 1 2
next
end

```

3. Configure VPN IPSEC Tunnel using SD-WAN control traffic

At The Data Center

a. Create VPN IPSEC Tunnel

HQ-A # config vpn ipsec phase1-interface

```

edit "VPN_to_BR_port2"
  set interface "port3"
  set peertype any
  set net-device disable
  set proposal des-md5 des-sha1

```

```
    set comments "VPN_to_BR_port2"
    set nattraversal disable
    set remote-gw 172.19.101.30
    set psksecret 123456
  next
end
HQ-A # config vpn ipsec phase2-interface
edit "VPN_to_BR_port2"
    set phase1name "VPN_to_BR_port2"
    set proposal des-md5 des-sha1
    set src-addr-type name
    set dst-addr-type name
    set src-name "VPN_via_BGP"
    set dst-name "VPN_via_BGP"
  next
end
HQ-A # config system interface
edit "VPN_to_BR_port2"
    set vdom "root"
    set ip 10.1.1.1 255.255.255.255
    set type tunnel
    set remote-ip 10.1.1.2 255.255.255.252
    set interface "port3"
  next
end
```

```
HQ-B # config vpn ipsec phase1-interface
edit "VPN_to_BR"
    set interface "port3"
    set peertype any
    set net-device disable
    set proposal des-md5 des-sha1
    set nattraversal disable
    set remote-gw 172.19.102.30
    set psksecret 123456
  next
end
HQ-B # config vpn ipsec phase2-interface
edit "VPN_to_BR"
    set phase1name "VPN_to_BR"
    set proposal des-md5 des-sha1
    set src-addr-type name
    set dst-addr-type name
    set src-name "VPN_via_BGP"
```

```

    set dst-name "VPN_via_BGP"
  next
end
HQ-B # config system interface
edit "VPN_to_BR"
  set vdom "root"
  set ip 10.1.1.5 255.255.255.255
  set type tunnel
  set remote-ip 10.1.1.6 255.255.255.252
  set interface "port3"
next
end

```

b. Configure eBGP routing with Branch Firewall

```

HQ-A # config router bgp
config neighbor
  edit "10.1.1.2"
    set soft-reconfiguration enable
    set remote-as 65120
  next
end
config network
  edit 1
    set prefix 10.92.101.0 255.255.255.0
  next
end

```

```

HQ-B # config router bgp
config neighbor
  edit "10.1.1.6"
    set soft-reconfiguration enable
    set remote-as 65120
  next
end
config network
  edit 1
    set prefix 10.92.102.0 255.255.255.0
  next
end

```

c. Create new SD-WAN Zone name VPN and assign SD-WAN members into this Zone

HQ-A # config system sdwan

```
config zone
  edit "VPN"
  next
end
config members
  edit 2
    set interface "VPN_to_BR_port2"
    set zone "VPN"
    set source 10.92.101.10
  next
end
```

HQ-B # config system sdwan

```
config zone
  edit "VPN"
  next
end
config members
  edit 2
    set interface "VPN_to_BR"
    set zone "VPN"
    set source 10.92.102.20
  next
end
```

d. Create Performance SLA to health check the internal IP address via VPN Link

HQ-A # config system sdwan

```
edit "CheckVPN"
  set server "10.159.100.254"
  set members 2
  config sla
    edit 1
      set latency-threshold 20
      set jitter-threshold 20
    next
  end
next
end
```

HQ-B # config system sdwan

```
edit "CheckVPN"
```

```

set server "10.159.100.254"
set members 2
config sla
  edit 1
    set latency-threshold 20
    set jitter-threshold 20
  next
end
next
end

```

e. Configure Firewall Policies for VPN Traffic and SD-WAN Rules for many types of Traffic via VPN tunnel link

HQ-A # config firewall policy

```

edit 2
  set name "VPN_IN"
  set srcintf "VPN"
  set dstintf "port2"
  set srcaddr "LAN-BR"
  set dstaddr "LAN-HQ"
  set action accept
  set schedule "always"
  set service "ALL"
  set ssl-ssh-profile "custom-deep-inspection"
next
edit 3
  set name "VPN_OUT"
  set srcintf "port2"
  set dstintf "VPN"
  set srcaddr "LAN-HQ"
  set dstaddr "LAN-BR"
  set action accept
  set schedule "always"
  set service "ALL"
  set ssl-ssh-profile "custom-deep-inspection"
next
end

```

HQ-A # config system sdwan

```

config service
  edit 7
    set name "VPN_SCTP"
    set mode sla

```

```
set protocol 132
set dst "LAN-BR"
set src "LAN-HQ"
config sla
  edit "CheckVPN"
    set id 1
  next
end
set priority-members 2
next
edit 3
  set name "VPN_CAPWAP"
  set mode sla
  set protocol 6
  set start-port 5246
  set end-port 5247
  set dst "LAN-BR"
  set src "LAN-HQ"
  config sla
    edit "CheckVPN"
      set id 1
    next
  end
  set priority-members 2
next
edit 4
  set name "VPN_DNS"
  set mode sla
  set protocol 6
  set start-port 53
  set end-port 53
  set dst "LAN-BR"
  set src "LAN-HQ"
  config sla
    edit "CheckVPN"
      set id 1
    next
  end
  set priority-members 2
next
edit 5
  set name "VPN_DHCP"
  set mode sla
```

```
set protocol 17
set start-port 67
set end-port 67
set dst "LAN-BR"
set src "LAN-HQ"
config sla
    edit "CheckVPN"
        set id 1
    next
end
set priority-members 2
next
edit 6
    set name "VPN_RADIUS"
    set mode sla
    set protocol 6
    set start-port 1812
    set end-port 1813
    set dst "LAN-BR"
    set src "LAN-HQ"
    config sla
        edit "CheckVPN"
            set id 1
        next
    end
    set priority-members 2
next
edit 1
    set name "VPN"
    set mode sla
    set dst "LAN-BR"
    set src "LAN-HQ"
    config sla
        edit "CheckVPN"
            set id 1
        next
    end
    set priority-members 2
next
end
```

HQ-B # config firewall policy

```
edit 2
    set name "VPN_IN"
```

```

set srcintf "VPN"
set dstintf "port2"
set srcaddr "LAN-BR"
set dstaddr "LAN-HQ"
set action accept
set schedule "always"
set service "ALL"
set ssl-ssh-profile "custom-deep-inspection"
next
edit 3
  set name "VPN_OUT"
  set srcintf "port2"
  set dstintf "VPN"
  set srcaddr "LAN-HQ"
  set dstaddr "LAN-BR"
  set action accept
  set schedule "always"
  set service "ALL"
  set ssl-ssh-profile "custom-deep-inspection"
next
end
HQ-B # config system sdwan
config service
  edit 3
    set name "VPN_SCTP"
    set mode sla
    set protocol 132
    set dst "LAN-BR"
    set src "LAN-HQ"
    config sla
      edit "CheckVPN"
        set id 1
      next
    end
    set priority-members 2
  next
  edit 4
    set name "VPN_DHCP"
    set mode sla
    set protocol 17
    set start-port 67
    set end-port 67
    set dst "LAN-BR"

```

```
set src "LAN-HQ"
config sla
  edit "CheckVPN"
    set id 1
  next
end
set priority-members 2
next
edit 5
  set name "VPN_DNS"
  set mode sla
  set protocol 17
  set start-port 53
  set end-port 53
  set dst "LAN-BR"
  set src "LAN-HQ"
  config sla
    edit "CheckVPN"
      set id 1
    next
  end
  set priority-members 2
next
edit 6
  set name "VPN_CAPWAP"
  set mode sla
  set protocol 6
  set start-port 5246
  set end-port 5247
  set dst "LAN-BR"
  set src "LAN-HQ"
  config sla
    edit "CheckVPN"
      set id 1
    next
  end
  set priority-members 2
next
edit 7
  set name "VPN_RADIUS"
  set mode sla
  set protocol 6
  set start-port 1812
```

```

set end-port 1813
set dst "LAN-BR"
set src "LAN-HQ"
config sla
    edit "CheckVPN"
        set id 1
    next
end
set priority-members 2
next
edit 1
    set name "VPN"
    set mode sla
    set dst "LAN-BR"
    set src "LAN-HQ"
    config sla
        edit "CheckVPN"
            set id 1
        next
    end
    set priority-members 2
next
end

```

- f. Configure the route-map under BGP routing to update the local-preference depending on community values received**

```

HQ-A # config router route-map
edit "comm1"
    config rule
        edit 1
            set match-community "20:1"
            set set-local-preference 200
        next
        edit 2
            set match-community "20:5"
            set set-local-preference 150
        next
    end
next
end
HQ-A # config router bgp
config neighbor

```

```

edit "10.1.1.2"
  set route-map-in "comm1"
next
end

```

HQ-B # config router route-map

```

edit "comm2"
  config rule
    edit 1
      set match-community "20:2"
      set set-local-preference 200
    next
    edit 2
      set match-community "20:5"
      set set-local-preference 150
    next
  end
next
end

```

HQ-B # config router bgp

```

config neighbor
  edit "10.1.1.6"
    set route-map-in "comm2"
  next
end

```

At Branch

a. Create VPN IPSEC Tunnels

Campus-Master # config vpn ipsec phase1-interface

```

edit "VPN_to_A_ISP1"
  set interface "port2"
  set peertype any
  set net-device disable
  set proposal des-md5 des-sha1
  set comments "VPN_to_A_ISP1"
  set nattraversal disable
  set remote-gw 172.18.101.10
  set psksecret 123456
next
edit "VPN_to_B_ISP2"
  set interface "port4"
  set peertype any

```

```

    set net-device disable
    set proposal des-md5 des-sha1
    set natTraversal disable
    set remote-gw 172.18.102.20
    set psksecret 123456
  next
end
Campus-Master # config vpn ipsec phase2-interface
edit "VPN_to_A_ISP1"
  set phase1name "VPN_to_A_ISP1"
  set proposal des-md5 des-sha1
  set src-addr-type name
  set dst-addr-type name
  set src-name "VPN_via_BGP"
  set dst-name "VPN_via_BGP"
next
edit "VPN_to_B_ISP2"
  set phase1name "VPN_to_B_ISP2"
  set proposal des-md5 des-sha1
  set src-addr-type name
  set dst-addr-type name
  set src-name "VPN_via_BGP"
  set dst-name "VPN_via_BGP"
next
end
Campus-Master # config system interface
edit "VPN_to_A_ISP1"
  set vdom "root"
  set ip 10.1.1.2 255.255.255.255
  set type tunnel
  set remote-ip 10.1.1.1 255.255.255.252
  set interface "port2"
next
edit "VPN_to_B_ISP2"
  set vdom "root"
  set ip 10.1.1.6 255.255.255.255
  set type tunnel
  set remote-ip 10.1.1.5 255.255.255.252
  set interface "port4"
next

```

b. Configure eBGP routing with two Firewall at the Data Center

Campus-Master # config router access-list

```
edit "net10.159"  
  config rule  
    edit 1  
      set prefix 10.159.100.0 255.255.255.0  
    next  
  end  
next  
end
```

Campus-Master # config router route-map

```
edit "comm1"  
  config rule  
    edit 1  
      set match-ip-address "net10.159"  
      set set-community "20:1"  
    next  
  end  
next  
edit "comm2"  
  config rule  
    edit 1  
      set match-ip-address "net10.159"  
      set set-community "20:2"  
    next  
  end  
next  
edit "comm5"  
  config rule  
    edit 1  
      set match-ip-address "net10.159"  
      set set-community "20:5"  
    next  
  end  
next  
end
```

Campus-Master # config router bgp

```
set as 65120  
set ebgp-multipath enable  
config neighbor  
  edit "10.1.1.1"  
    set soft-reconfiguration enable  
    set remote-as 65100  
    set route-map-out "comm5"
```

```

    set route-map-out-preferable "comm1"
  next
  edit "10.1.1.5"
    set soft-reconfiguration enable
    set remote-as 65100
    set route-map-out "comm5"
    set route-map-out-preferable "comm2"
  next
end
config network
  edit 1
    set prefix 10.159.100.0 255.255.255.0
  next
end

```

c. Create new SD-WAN Zone name VPN and assign SD-WAN members into this Zone

Campus-Master # config system sdwan

```

config zone
  edit "VPN"
  next
end
config members
  edit 3
    set interface "VPN_to_A_ISP1"
    set zone "VPN"
    set source 10.159.100.254
    set cost 5
  next
  edit 4
    set interface "VPN_to_B_ISP2"
    set zone "VPN"
    set source 10.159.100.254
    set cost 10
  next
end

```

d. Create Performance SLA to health check the internal IP address via VPN Links

Campus-Master # config system sdwan

```

config health-check

```

```

edit "CheckVPN_A"
  set server "10.92.101.10"
  set members 3
  config sla
    edit 1
      set latency-threshold 20
      set jitter-threshold 20
    next
  end
next
edit "CheckVPN_B"
  set server "10.92.102.20"
  set members 4
  config sla
    edit 1
      set latency-threshold 20
      set jitter-threshold 20
    next
  end
next
end
Campus-Master # config system sdwan
config neighbor
  edit "10.1.1.1"
    set member 3
    set role primary
    set health-check "CheckVPN_A"
    set sla-id 1
  next
  edit "10.1.1.5"
    set member 4
    set role secondary
    set health-check "CheckVPN_B"
    set sla-id 1
  next
end

```

e. Configure Firewall Policies for VPN Traffic and SD-WAN Rules for many types of Traffic via VPN tunnel links

```

Campus-Master # config system sdwan

```

```

config service
  edit 11

```

```
set name "VPN_SCTP"
set mode sla
set role primary
set protocol 132
set dst "LAN-HQ"
set src "LAN-BR"
config sla
  edit "CheckVPN_A"
    set id 1
  next
end
set priority-members 3
next
edit 12
  set name "VPN_SCTP_BK"
  set mode sla
  set role secondary
  set protocol 132
  set dst "LAN-HQ"
  set src "LAN-BR"
  config sla
    edit "CheckVPN_B"
      set id 1
    next
  end
  set priority-members 4
next
edit 1
  set name "VPN_CAPWAP"
  set mode sla
  set role primary
  set protocol 6
  set start-port 5246
  set end-port 5247
  set dst "LAN-HQ"
  set src "LAN-BR"
  config sla
    edit "CheckVPN_A"
      set id 1
    next
  end
  set priority-members 3
next
```

```
edit 4
  set name "VPN_CAPWAP_BK"
  set mode sla
  set role secondary
  set protocol 6
  set start-port 5246
  set end-port 5247
  set dst "LAN-HQ"
  set src "LAN-BR"
  config sla
    edit "CheckVPN_B"
      set id 1
    next
  end
  set priority-members 4
next
edit 5
  set name "VPN_DNS"
  set mode sla
  set role primary
  set protocol 6
  set start-port 53
  set end-port 53
  set dst "LAN-HQ"
  set src "LAN-BR"
  config sla
    edit "CheckVPN_A"
      set id 1
    next
  end
  set priority-members 3
next
edit 6
  set name "VPN_DNS_BK"
  set mode sla
  set role secondary
  set protocol 6
  set start-port 53
  set end-port 53
  set dst "LAN-HQ"
  set src "LAN-BR"
  config sla
    edit "CheckVPN_B"
```

```
        set id 1
      next
    end
    set priority-members 4
  next
edit 7
  set name "VPN_RADIUS"
  set mode sla
  set role primary
  set protocol 6
  set start-port 1812
  set end-port 1813
  set dst "LAN-HQ"
  set src "LAN-BR"
  config sla
    edit "CheckVPN_A"
      set id 1
    next
  end
  set priority-members 3
next
edit 8
  set name "VPN_RADIUS_BK"
  set mode sla
  set role secondary
  set dst "LAN-HQ"
  set src "LAN-BR"
  config sla
    edit "CheckVPN_B"
      set id 1
    next
  end
  set priority-members 4
next
edit 9
  set name "VPN_DCHP"
  set mode sla
  set role primary
  set protocol 17
  set start-port 67
  set end-port 67
  set dst "LAN-HQ"
  set src "LAN-BR"
```

```

config sla
    edit "CheckVPN_A"
        set id 1
    next
end
set priority-members 3
next
edit 10
    set name "VPN_DHCP_BK"
    set mode sla
    set role secondary
    set protocol 17
    set start-port 67
    set end-port 67
    set dst "LAN-HQ"
    set src "LAN-BR"
    config sla
        edit "CheckVPN_B"
            set id 1
        next
    end
    set priority-members 4
next
edit 3
    set name "VPN"
    set mode load-balance
    set dst "LAN-HQ"
    set src "LAN-BR"
    config sla
        edit "CheckVPN_A"
            set id 1
        next
        edit "CheckVPN_B"
            set id 1
        next
    end
    set priority-members 4 3
next

```

4. Marking DSCP for the outgoing traffic and apply the Traffic Shaping Profile under WAN interfaces

At The Data Center

a. Marking DSCP for the outgoing traffic

HQ-A # config system sdwan

```
config service
  edit 7
    set name "VPN_SCTP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 3
    set name "VPN_CAPWAP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 4
    set name "VPN_DNS"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 5
    set name "VPN_DHCP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 6
    set name "VPN_RADIUS"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 1
    set name "VPN"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 100010
```

```
    set dscp-reverse-tag 100010
  next
end
```

HQ-B # config system sdwan

```
config service
  edit 3
    set name "VPN_SCTP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 4
    set name "VPN_DHCP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 5
    set name "VPN_DNS"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 6
    set name "VPN_CAPWAP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 7
    set name "VPN_RADIUS"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 1
    set name "VPN"
    set dscp-forward enable
```

```

set dscp-reverse enable
set dscp-forward-tag 100010
set dscp-reverse-tag 100010
next

```

b. Create the Traffic Shaping Policy and Profile

HQ-A # *config firewall shaping-policy*

```

edit 1
  set name "Critical_Apps"
  set service "ALL_ICMP" "BGP" "DHCP" "DNS" "RADIUS" "SMTP" "CAPWAP" "SCTP"
  set dstintf "VPN"
  set traffic-shaper "high-priority"
  set traffic-shaper-reverse "high-priority"
  set srcaddr "TUNNEL_ADD" "LAN-HQ"
  set dstaddr "LAN-BR" "TUNNEL_ADD"
next
edit 2
  set name "Business_Apps"
  set service "ALL"
  set dstintf "VPN"
  set traffic-shaper "medium-priority"
  set traffic-shaper-reverse "medium-priority"
  set srcaddr "LAN-HQ"
  set dstaddr "LAN-BR"
next
edit 3
  set name "Secure_Internet"
  set service "ALL"
  set dstintf "INTERNET"
  set traffic-shaper "low-priority"
  set traffic-shaper-reverse "low-priority"
  set srcaddr "all"
  set dstaddr "all"
next
end

```

HQ-A # *config firewall shaping-profile*

```

edit "All_Service"
  set default-class-id 4
  config shaping-entries
    edit 1
      set class-id 4
      set priority low
    next
  next
end

```

```

        set guaranteed-bandwidth-percentage 2
        set maximum-bandwidth-percentage 100
    next
    edit 2
        set class-id 2
        set priority critical
        set guaranteed-bandwidth-percentage 90
        set maximum-bandwidth-percentage 100
    next
    edit 3
        set class-id 3
        set guaranteed-bandwidth-percentage 8
        set maximum-bandwidth-percentage 100
    next
end
next
end

```

HQ-B # *config firewall shaping-policy*

```

edit 1
    set name "Critical_Apps"
    set service "ALL_ICMP" "BGP" "DHCP" "DNS" "RADIUS" "SMTP" "CAPWAP" "SCTP"
    set dstintf "VPN"
    set traffic-shaper "high-priority"
    set traffic-shaper-reverse "high-priority"
    set srcaddr "TUNNEL_ADD" "LAN-HQ"
    set dstaddr "LAN-BR" "TUNNEL_ADD"
next
edit 2
    set name "Business_Apps"
    set service "ALL"
    set dstintf "VPN"
    set traffic-shaper "medium-priority"
    set traffic-shaper-reverse "medium-priority"
    set srcaddr "LAN-HQ"
    set dstaddr "LAN-BR"
next
edit 3
    set name "Secure_Internet"
    set service "ALL"
    set dstintf "INTERNET"
    set traffic-shaper "low-priority"
    set traffic-shaper-reverse "low-priority"
    set srcaddr "all"

```

```

    set dstaddr "all"
  next
end
HQ-B # config firewall shaping-profile
edit "All_Service"
  set default-class-id 4
  config shaping-entries
    edit 1
      set class-id 2
      set priority critical
      set guaranteed-bandwidth-percentage 90
      set maximum-bandwidth-percentage 100
    next
    edit 2
      set class-id 3
      set guaranteed-bandwidth-percentage 8
      set maximum-bandwidth-percentage 100
    next
    edit 3
      set class-id 4
      set priority low
      set guaranteed-bandwidth-percentage 2
      set maximum-bandwidth-percentage 100
    next
  end
next
end

```

c. Apply the Traffic Shaping Profile under WAN interfaces

HQ-A # config system interface

```

edit "port3"
  set outbandwidth 2000
  set egress-shaping-profile "All_Service"
next

```

HQ-B # config system interface

```

edit "port3"
  set outbandwidth 2000
  set egress-shaping-profile "All_Service"
next

```

At Branch

a. Marking DSCP for the outgoing traffic

Campus-Master # config system sdwan

```
config service
  edit 11
    set name "VPN_SCTP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 12
    set name "VPN_SCTP_BK"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 1
    set name "VPN_CAPWAP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 4
    set name "VPN_CAPWAP_BK"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 5
    set name "VPN_DNS"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
  next
  edit 6
    set name "VPN_DNS_BK"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
```

```
    set dscp-reverse-tag 101110
next
edit 7
    set name "VPN_RADIUS"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
next
edit 8
    set name "VPN_RADIUS_BK"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
next
edit 9
    set name "VPN_DCHP"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
next
edit 10
    set name "VPN_DHCP_BK"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 101110
    set dscp-reverse-tag 101110
next
edit 3
    set name "VPN"
    set dscp-forward enable
    set dscp-reverse enable
    set dscp-forward-tag 100010
    set dscp-reverse-tag 100010
next
```

b. Create the Traffic Shaping Policy and Profile

```
Campus-Master # config firewall shaping-policy
```

```
edit 1
    set name "Critical_Apps"
```

```

set service "BGP" "DHCP" "DNS" "RADIUS" "SMTP" "Windows AD" "CAPWAP" "SCTP"
set dstintf "VPN"
set class-id 2
set srcaddr "LAN-BR" "TUNNEL_ADD"
set dstaddr "TUNNEL_ADD" "LAN-HQ"
next
edit 2
set name "Business_Apps"
set service "ALL"
set dstintf "VPN"
set class-id 3
set srcaddr "LAN-BR"
set dstaddr "LAN-HQ"
next
edit 3
set name "Secure_Internet"
set service "ALL"
set dstintf "INTERNET"
set class-id 4
set srcaddr "all"
set dstaddr "all"
next
end
Campus-Master # config firewall shaping-profile
edit "All_Service"
set default-class-id 4
config shaping-entries
edit 1
set class-id 2
set priority critical
set guaranteed-bandwidth-percentage 90
set maximum-bandwidth-percentage 100
next
edit 2
set class-id 3
set guaranteed-bandwidth-percentage 8
set maximum-bandwidth-percentage 100
next
edit 3
set class-id 4
set priority low
set guaranteed-bandwidth-percentage 2
set maximum-bandwidth-percentage 100

```

```

    next
  end
next
end

```

c. Apply the Traffic Shaping Profile under WAN interfaces

Campus-Master # config system interface

```

edit "port2"
  set outbandwidth 2000
  set egress-shaping-profile "All_Service"
next
edit "port4"
  set outbandwidth 2000
  set egress-shaping-profile "All_Service"
next

```

Verification:

1. Verification the Critical traffic will be transferred on the primary IPSEC tunnel link when it meets the SLA threshold

- Check selected neighbor on Branch Firewall

```

Campus-Master # diagnose sys sdwan neighbor
SD-WAN neighbor status: hold-down(disable), hold-down-time(0), hold_boot_time(0)
    Selected role(primary) last_secondary_select_time/current_time in seconds 0/12795
Neighbor(10.1.1.1): member(3) role(primary)
    Health-check(CheckVPN_A:1) sla-pass selected alive
Neighbor(10.1.1.5): member(4) role(secondary)
    Health-check(CheckVPN_B:1) sla-pass alive

```

- In the normal state, two VPN IPSEC links all get SLA target

```

Campus-Master # diagnose sys sdwan health-check
Health Check(Default_DNS):
Seq(1 port2): state(alive), packet-loss(1.000%) latency(100.540), jitter(2.785) sla_map=0x1
Seq(2 port4): state(alive), packet-loss(1.000%) latency(101.101), jitter(3.349) sla_map=0x0
Health Check(CheckINTERNET):
Seq(1 port2): state(alive), packet-loss(0.000%) latency(31.964), jitter(2.909) sla_map=0x1
Seq(2 port4): state(alive), packet-loss(0.000%) latency(32.985), jitter(3.966) sla_map=0x1
Health Check(CheckVPN_A):
Seq(3 VPN_to_A_ISP1): state(alive), packet-loss(0.000%) latency(2.443), jitter(0.599) sla_map=0x1
Health Check(CheckVPN_B):
Seq(4 VPN_to_B_ISP2): state(alive), packet-loss(0.000%) latency(2.456), jitter(0.843) sla_map=0x1

```

- Check BGP community that Branch Firewall advertised to two neighbors at the Data Center

```

HQ-A # get router info bgp network 10.159.100.0
VRF 0 BGP routing table entry for 10.159.100.0/24
Paths: (1 available, best #1, table Default-IP-Routing-Table)
  Advertised to non peer-group peers:
    10.92.101.252
  Original VRF 0
  65120
    10.1.1.2 from 10.1.1.2 (192.168.74.30)
      Origin IGP metric 0, localpref 200, valid, external, best
      Community: 20:1
      Last update: Sat Dec 19 23:12:33 2020

```

```

HQ-B # get router info bgp network 10.159.100.0
VRF 0 BGP routing table entry for 10.159.100.0/24
Paths: (2 available, best #2, table Default-IP-Routing-Table)
  Advertised to non peer-group peers:
    10.1.1.6
  Original VRF 0
  65120
    10.1.1.6 from 10.1.1.6 (192.168.74.30)
      Origin IGP metric 0, localpref 150, valid, external
      Community: 20:5
      Last update: Sat Dec 19 23:20:21 2020

```

-> On two Firewalls at the Data Center depending on the community that they received from the Branch Firewall will advertise the local-preference to Router Backbone. (200 for community 20:1 and 150 for community 20:5)

- On Router Backbone, the best path to Firewall HQ-A

```

BB-A#show ip bgp
BGP table version is 53, local router ID is 10.92.92.252
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network                Next Hop              Metric LocPrf Weight Path
*>i 0.0.0.0                10.92.101.10          0      120      0 ?
* i 10.84.84.0/24          10.92.92.253          0      100      0 i
*>                          0.0.0.0               0           32768 i
* i 10.84.100.0/24         10.92.92.253          0      100      0 i
*>                          0.0.0.0               0           32768 i
* i 10.92.92.0/30          10.92.92.253          0      100      0 i
*>                          0.0.0.0               0           32768 i
*> 10.92.92.252/32         0.0.0.0               0           32768 i
r>i 10.92.92.253/32        10.92.92.253          0      100      0 i
* i 10.92.101.0/24         10.92.101.10          0      100      0 i
*>                          0.0.0.0               0           32768 i
*>i 10.92.102.0/24         10.92.92.253          0      100      0 i
*>i 10.159.100.0/24        10.92.101.10          0      200      0 65120 i

```

- Install nmap on USR PC at Branch and on Server PC at the Data Center.
- Generate the traffic as DHCP, DNS, CAPWAP, RADIUS, SMTP, SCTP to test traffic via SD-WAN rules
 - Nping --udp -p 67 10.84.84.150
 - Nping --udp -p 53 10.84.84.150
 - Nping --tcp -p 5246 10.84.84.150
 - Nping --tcp -p 1812 10.84.84.150
 - Nping --tcp -p 25 10.84.84.150
 - Nmap -sY 10.84.84.150 *//For SCTP INIT*
- On USR PC:

```

C:\>nping --tcp -p 5246 10.84.84.150
Starting Nping 0.7.80 < https://nmap.org/nping > at 2020-12-19 13:44 SE Asia Standard Time
SENT (0.2490s) TCP 10.159.100.1:22504 > 10.84.84.150:5246 S ttl=64 id=27902 iplen=40 seq=624599294 win=1480
RCVD (0.2490s) TCP 10.84.84.150:5246 > 10.159.100.1:22504 RA ttl=125 id=5 iplen=40 seq=0 win=0
SENT (1.2630s) TCP 10.159.100.1:22504 > 10.84.84.150:5246 S ttl=64 id=27902 iplen=40 seq=624599294 win=1480
RCVD (1.2630s) TCP 10.84.84.150:5246 > 10.159.100.1:22504 RA ttl=125 id=6 iplen=40 seq=0 win=0
SENT (2.2770s) TCP 10.159.100.1:22504 > 10.84.84.150:5246 S ttl=64 id=27902 iplen=40 seq=624599294 win=1480
RCVD (2.2770s) TCP 10.84.84.150:5246 > 10.159.100.1:22504 RA ttl=125 id=7 iplen=40 seq=0 win=0
SENT (3.2910s) TCP 10.159.100.1:22504 > 10.84.84.150:5246 S ttl=64 id=27902 iplen=40 seq=624599294 win=1480
RCVD (3.2910s) TCP 10.84.84.150:5246 > 10.159.100.1:22504 RA ttl=125 id=8 iplen=40 seq=0 win=0
SENT (4.3050s) TCP 10.159.100.1:22504 > 10.84.84.150:5246 S ttl=64 id=27902 iplen=40 seq=624599294 win=1480
RCVD (4.3050s) TCP 10.84.84.150:5246 > 10.159.100.1:22504 RA ttl=125 id=9 iplen=40 seq=0 win=0
Max rtt: 0.000ms ! Min rtt: 0.000ms ! Avg rtt: 0.000ms
Raw packets sent: 5 (270B) ! Rcvd: 5 (230B) ! Lost: 0 (0.00%)
Nping done: 1 IP address pinged in 4.30 seconds
C:\>nmap -sV 10.84.84.150
Starting Nmap 7.80 < https://nmap.org > at 2020-12-19 13:44 SE Asia Standard Time
Nmap scan report for 10.84.84.150
Host is up (0.015s latency).
All 52 scanned ports on 10.84.84.150 are filtered
Nmap done: 1 IP address (1 host up) scanned in 13.42 seconds
C:\>_

```

- Check Hit Count on SD-WAN Rules on the Branch Firewall

IPv4 12						
11	VPN_SCTP	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
12	VPN_SCTP_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
1	VPN_CAPWAP	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	4,295
4	VPN_CAPWAP_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
5	VPN_DNS	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
6	VPN_DNS_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
7	VPN_RADIUS	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
8	VPN_RADIUS_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
9	VPN_DCHP	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
10	VPN_DHCP_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
3	VPN	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2 VPN_to_A_ISP1	912
2	INTERNET	all	all	SLA	WAN1 (port2) WAN2 (port4)	236

2. Verification failover when the primary Tunnel link does not meet the SLA threshold

- Set the SLA threshold decrease to test

Name	Detect Server	Packet Loss	Latency	Jitter	Failure Threshold	Recc
CheckINTERNET	8.8.8.8	WAN1 (port2): 0.00% WAN2 (port4): 0.00%	WAN1 (port2): 31.22ms WAN2 (port4): 32.10ms	WAN1 (port2): 1.17ms WAN2 (port4): 2.30ms	5	5
CheckVPN_A	10.92.101.10	VPN_to_A_ISP1: 0.00%	VPN_to_A_ISP1: 3.23ms	VPN_to_A_ISP1: 1.33ms	5	5
CheckVPN_B	10.92.102.20	VPN_to_B_ISP2: 0.00%	VPN_to_B_ISP2: 3.42ms	VPN_to_B_ISP2: 1.79ms	5	5

- Check the SD-WAN neighbor status on Branch Firewall

```
Campus-Master # diagnose sys sdwan neighbor
SD-WAN neighbor status: hold-down(disable), hold-down-time(0), hold_boot_time(0)
Selected role(secondary) last_secondary_select_time/current_time in seconds 14118/14119
Neighbor(10.1.1.1): member(3) role(primary)
Health-check(CheckVPN_A:1) sla-fail alive
Neighbor(10.1.1.5): member(4) role(secondary)
Health-check(CheckVPN_B:1) sla-pass selected alive
```

- Check the BGP community updated to BGP neighbors at the Data Center.

```
HQ-A # get router info bgp network 10.159.100.0
VRF 0 BGP routing table entry for 10.159.100.0/24
Paths: (2 available, best #1, table Default-IP-Routing-Table)
Advertised to non peer-group peers:
10.1.1.2
Original VRF 0
65120
10.92.102.20 from 10.92.101.252 (192.168.74.20)
Origin IGP metric 0, localpref 200, valid, internal, best
Community: 20:2
Originator: 192.168.74.20, Cluster list: 10.92.92.252 10.92.92.253
Last update: Sat Dec 19 23:45:08 2020

Original VRF 0
65120
10.1.1.2 from 10.1.1.2 (192.168.74.30)
Origin IGP metric 0, localpref 150, valid, external
Community: 20:5
Last update: Sat Dec 19 23:44:38 2020
```

-> On Firewall HQ-A received community 20:5 from the Branch Firewall and it will advertise this prefix with the local-preference 150

```

HQ-B # get router info bgp network 10.159.100.0
VRF 0 BGP routing table entry for 10.159.100.0/24
Paths: (1 available, best #1, table Default-IP-Routing-Table)
  Advertised to non peer-group peers:
    10.92.102.253
  Original VRF 0
  65120
    10.1.1.6 from 10.1.1.6 (192.168.74.30)
      Origin IGP metric 0, localpref 200, valid, external, best
      Community: 20:2
      Last update: Sat Dec 19 23:44:42 2020

```

-> On Firewall HQ-B received community 20:2, it will advertise with the local-preference 200 to Router Backbone

- On Router Backbone, the best path to Firewall HQ-B

```

BB-A#show ip bgp
BGP table version is 56, local router ID is 10.92.92.252
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network                Next Hop                Metric LocPrf Weight Path
*>i 0.0.0.0                10.92.101.10              0      120      0 ?
* i 10.84.84.0/24          10.92.92.253              0      100      0 i
*>                          0.0.0.0                  0           32768 i
* i 10.84.100.0/24         10.92.92.253              0      100      0 i
*>                          0.0.0.0                  0           32768 i
* i 10.92.92.0/30          10.92.92.253              0      100      0 i
*>                          0.0.0.0                  0           32768 i
*> 10.92.92.252/32         0.0.0.0                  0           32768 i
r>i 10.92.92.253/32        10.92.92.253              0      100      0 i
* i 10.92.101.0/24         10.92.101.10              0      100      0 i
*>                          0.0.0.0                  0           32768 i
*>i 10.92.102.0/24         10.92.92.253              0      100      0 i
*>i 10.159.100.0/24        10.92.102.20              0      200      0 65120 i

```

- Ping Test and Generate CAPWAP traffic from USR PC to Server PC at the Data Center
- On USR PC

```
C:\Users>ping 10.84.84.150
```

```
Pinging 10.84.84.150 with 32 bytes of data:
Reply from 10.84.84.150: bytes=32 time=22ms TTL=124
Reply from 10.84.84.150: bytes=32 time=6ms TTL=124
Reply from 10.84.84.150: bytes=32 time=5ms TTL=124
Reply from 10.84.84.150: bytes=32 time=5ms TTL=124

Ping statistics for 10.84.84.150:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 22ms, Average = 9ms
```

```
C:\Users>nping --tcp -p 5246 -c 10 10.84.84.150
```

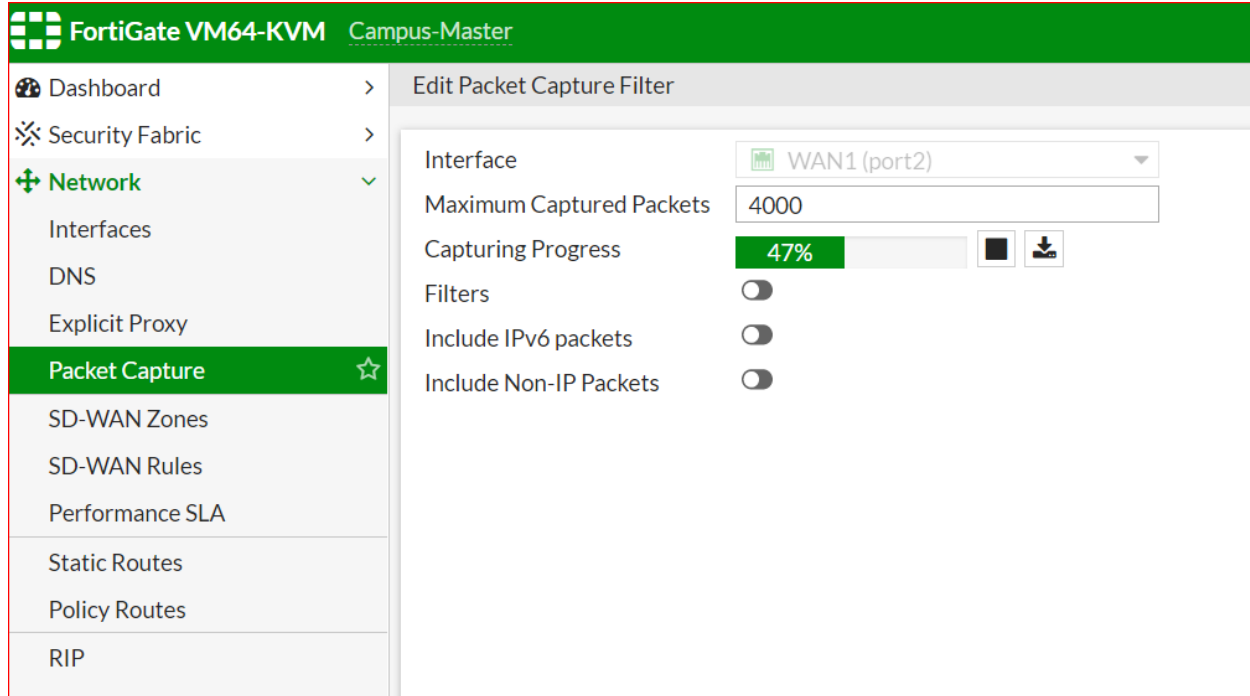
```
Starting Nping 0.7.80 ( https://nmap.org/nping ) at 2020-12-20 07:54 SE Asia Standard Time
SENT (0.2970s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (0.2970s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8279 ip len=40 seq=0 win=0
SENT (1.3110s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (1.3110s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8280 ip len=40 seq=0 win=0
SENT (2.3250s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (2.3250s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8281 ip len=40 seq=0 win=0
SENT (3.3390s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (3.3390s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8285 ip len=40 seq=0 win=0
SENT (4.3530s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (4.3530s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8284 ip len=40 seq=0 win=0
SENT (5.3670s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (5.3670s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8286 ip len=40 seq=0 win=0
SENT (6.3810s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (6.3810s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8286 ip len=40 seq=0 win=0
SENT (7.3950s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (7.3950s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8286 ip len=40 seq=0 win=0
SENT (8.4090s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (8.4090s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8286 ip len=40 seq=0 win=0
SENT (9.4230s) TCP 10.159.100.1:31002 > 10.84.84.150:5246 S ttl=64 id=53148 ip len=40 seq=389746812 win=1480
RCVD (9.4230s) TCP 10.84.84.150:5246 > 10.159.100.1:31002 RA ttl=124 id=8286 ip len=40 seq=0 win=0
```

- Check Hit Count on SD-WAN Rule on the Branch Firewall

IPv4						
11	VPN_SCTP	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
12	VPN_SCTP_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
1	VPN_CAPWAP	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	4,295
4	VPN_CAPWAP_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	34
5	VPN_DNS	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
6	VPN_DNS_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
7	VPN_RADIUS	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
8	VPN_RADIUS_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	2
9	VPN_DCHP	LAN-BR	LAN-HQ	SLA	VPN_to_A_ISP1	0
10	VPN_DHCP_BK	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2	0
3	VPN	LAN-BR	LAN-HQ	SLA	VPN_to_B_ISP2 VPN_to_A_ISP1	923
2	INTERNET	all	all	SLA	WAN1 (port2) WAN2 (port4)	241

3. Verification DSCP mapping on ESP packets via ISP network.

- This link for how to decrypt ESP packets on FortiGate Firewall:
<https://kb.fortinet.com/kb/documentLink.do?externalID=FD48280>
- Generate the VPN traffic on USR PC to the Data Center and capture packets with on WAN interface



- Use wireshark to decrypt ESP packets

...	54.340771	10.159.100.1	10.84.84.150	TCP	110 61277 → 5246 [SYN] Seq=0 Win=1480 Len=0
...	55.351457	10.159.100.1	10.84.84.150	TCP	110 [TCP Retransmission] 61277 → 5246 [SYN] Seq=0 Win=1480 Len=0
<					
> Differentiated Services Field: 0xb8 (DSCP: EF PHB, ECN: Not-ECT)					
Total Length: 96					
Identification: 0x1135 (4405)					
> Flags: 0x0000					
Fragment offset: 0					
Time to live: 62					
Protocol: Encap Security Payload (50)					
Header checksum: 0x4831 [validation disabled]					
[Header checksum status: Unverified]					
Source: 172.19.101.30					
Destination: 172.18.101.10					
> Encapsulating Security Payload					
Internet Protocol Version 4, Src: 10.159.100.1, Dst: 10.84.84.150					
0100 = Version: 4					
... 0101 = Header Length: 20 bytes (5)					
> Differentiated Services Field: 0xb8 (DSCP: EF PHB, ECN: Not-ECT)					
Total Length: 40					
Identification: 0x2c94 (11412)					
> Flags: 0x0000					
Fragment offset: 0					
Time to live: 63					
Protocol: TCP (6)					
Header checksum: 0x80fa [validation disabled]					
[Header checksum status: Unverified]					
Source: 10.159.100.1					
Destination: 10.84.84.150					
Transmission Control Protocol, Src Port: 61277, Dst Port: 5246, Seq: 0, Len: 0					
Source Port: 61277					
Destination Port: 5246					
[Stream index: 2]					
[TCP Segment Len: 0]					
Sequence number: 0 (relative sequence number)					

4. Verification Traffic shaping on WAN interface

- Check allocated bandwidth on WAN1 interface (assume the link to ISP2 down -> all traffic via one physical link)

```
Campus-Master # diagnose netlink interface list port2

if=port2 family=00 type=1 index=4 mtu=1500 link=0 master=0
ref=22 state=start present fw_flags=10000000 flags=up broadcast run promsic multicast
Qdisc=pfifo_fast hw_addr=00:09:0f:09:00:01 broadcast_addr=ff:ff:ff:ff:ff:ff
egress traffic control:
    bandwidth=2000(kbps) lock_hit=0 default_class=4 n_active_class=3
    class-id=4      allocated-bandwidth=40(kbps)      guaranteed-bandwidth=40(kbps)
                   max-bandwidth=2000(kbps)          current-bandwidth=7(kbps)
                   priority=low      forwarded_bytes=90K
                   dropped_packets=0      dropped_bytes=0
    class-id=3      allocated-bandwidth=160(kbps)     guaranteed-bandwidth=160(kbps)
                   max-bandwidth=2000(kbps)          current-bandwidth=0(kbps)
                   priority=high     forwarded_bytes=0
                   dropped_packets=0      dropped_bytes=0
    class-id=2      allocated-bandwidth=1800(kbps)    guaranteed-bandwidth=1800(kbps)
                   max-bandwidth=2000(kbps)          current-bandwidth=0(kbps)
                   priority=critical  forwarded_bytes=0
                   dropped_packets=0      dropped_bytes=0
stat: rxp=1397 txp=1400 rxb=148286 txb=121544 rx=0 tx=0 rxd=0 txd=0 mc=0 collision=0
re: rxl=0 rxo=0 rxc=0 rxf=0 rxfi=0 rxm=0
te: txa=0 txc=0 txfi=0 txh=0 txw=0
misc rxc=0 txc=0
input_type=0 state=3 arp_entry=0 refcnt=22
```

- Now generate the Internet Traffic from PC to check allocated bandwidth in that time

```

Campus-Master # diagnose netlink interface list port2

if=port2 family=00 type=1 index=4 mtu=1500 link=0 master=0
ref=23 state=start present fw_flags=10000000 flags=up broadcast run promsic multicast
Qdisc=pfifo_fast hw_addr=00:09:0f:09:00:01 broadcast_addr=ff:ff:ff:ff:ff:ff
egress traffic control:
    bandwidth=2000(kbps) lock_hit=25 default_class=4 n_active_class=3
    class-id=4      allocated-bandwidth=540(kbps)   guaranteed-bandwidth=40(kbps)
                   max-bandwidth=2000(kbps)       current-bandwidth=430(kbps)
                   priority=low   forwarded_bytes=855K
                   dropped_packets=0   dropped_bytes=0
    class-id=3      allocated-bandwidth=20(kbps)   guaranteed-bandwidth=160(kbps)
                   max-bandwidth=2000(kbps)       current-bandwidth=0(kbps)
                   priority=high  forwarded_bytes=0
                   dropped_packets=0   dropped_bytes=0
    class-id=2      allocated-bandwidth=1440(kbps) guaranteed-bandwidth=1800(kbps)
                   max-bandwidth=2000(kbps)       current-bandwidth=0(kbps)
                   priority=critical forwarded_bytes=0
                   dropped_packets=0   dropped_bytes=0
stat: rxb=2284 txp=15015 rxb=242766 txb=885888 rx=0 tx=0 rxd=0 txd=0 mc=0 collision=0
re: rxl=0 rxo=0 rxc=0 rxf=0 rxfi=0 rxm=0
te: txa=0 txc=0 txfi=0 txh=0 txw=0
misc rxc=0 txc=0
input_type=0 state=3 arp_entry=0 refcnt=23

```

- Generate the VPN traffic between Branch and the Data Center (Business Apps) with bandwidth equal 1.7M

```

Campus-Master # diagnose netlink interface list port2

if=port2 family=00 type=1 index=4 mtu=1500 link=0 master=0
ref=22 state=start present fw_flags=10000000 flags=up broadcast run promsic multicast
Qdisc=pfifo_fast hw_addr=00:09:0f:09:00:01 broadcast_addr=ff:ff:ff:ff:ff:ff
egress traffic control:
    bandwidth=2000(kbps) lock_hit=865 default_class=4 n_active_class=3
    class-id=4      allocated-bandwidth=180(kbps)   guaranteed-bandwidth=40(kbps)
                   max-bandwidth=2000(kbps)       current-bandwidth=189(kbps)
                   priority=low   forwarded_bytes=3593K
                   dropped_packets=3957   dropped_bytes=215K
    class-id=3      allocated-bandwidth=1800(kbps) guaranteed-bandwidth=160(kbps)
                   max-bandwidth=2000(kbps)       current-bandwidth=1786(kbps)
                   priority=high  forwarded_bytes=4136K
                   dropped_packets=0   dropped_bytes=0
    class-id=2      allocated-bandwidth=20(kbps)   guaranteed-bandwidth=1800(kbps)
                   max-bandwidth=2000(kbps)       current-bandwidth=0(kbps)
                   priority=critical forwarded_bytes=0
                   dropped_packets=0   dropped_bytes=0
stat: rxb=838 txp=68988 rxb=88246 txb=7721007 rx=0 tx=0 rxd=0 txd=0 mc=0 collision=0
re: rxl=0 rxo=0 rxc=0 rxf=0 rxfi=0 rxm=0
te: txa=0 txc=0 txfi=0 txh=0 txw=0
misc rxc=0 txc=0
input_type=0 state=3 arp_entry=0 refcnt=22

```

-> *Secure_Internet Class Traffic will be dropped to give bandwidth for Business Class with the priority higher.*

- Continue generate DHCP traffic (UDP 67) from PC to the Data Center (Critical_Apps) with bandwidth equal 1.8M

```
Campus-Master # diagnose netlink interface list port2

if=port2 family=00 type=1 index=4 mtu=1500 link=0 master=0
ref=23 state=start present fw_flags=10000000 flags=up broadcast run promsic multicast
Qdisc=pfifo_fast hw_addr=00:09:0f:09:00:01 broadcast_addr=ff:ff:ff:ff:ff:ff
egress traffic control:
    bandwidth=2000(kbps) lock_hit=2586 default_class=4 n_active_class=3
    class-id=4      allocated-bandwidth=60(kbps)   guaranteed-bandwidth=40(kbps)
                  max-bandwidth=2000(kbps)      current-bandwidth=58(kbps)
                  priority=low   forwarded_bytes=7952K
                  dropped_packets=45K   dropped_bytes=2493K
    class-id=3      allocated-bandwidth=160(kbps)   guaranteed-bandwidth=160(kbps)
                  max-bandwidth=2000(kbps)      current-bandwidth=140(kbps)
                  priority=high  forwarded_bytes=39454K
                  dropped_packets=38   dropped_bytes=54K
    class-id=2      allocated-bandwidth=1780(kbps)   guaranteed-bandwidth=1800(kbps)
                  max-bandwidth=2000(kbps)      current-bandwidth=1731(kbps)
                  priority=critical forwarded_bytes=3676K
                  dropped_packets=16   dropped_bytes=19K
stat: rxp=2171 txp=176713 rxb=229529 txb=51083314 rxe=0 txe=0 rxd=0 txd=0 mc=0 collision=0
re: rxl=0 rxo=0 rxc=0 rxf=0 rxfi=0 rxm=0
te: txa=0 txc=0 txfi=0 txh=0 txw=0
misc rxc=0 txc=0
input_type=0 state=3 arp_entry=0 refcnt=23
```

-> *Current Bandwidth of Secure Internet and Business Classes have decreased to guaranteed bandwidth.*